



Explanation

Part 2

Carl G. Hempel (1905-1997)
Philosopher & logician
Formulated the D-N model

ANNOUNCEMENT

Assignment: Role Play of Scientific Figure

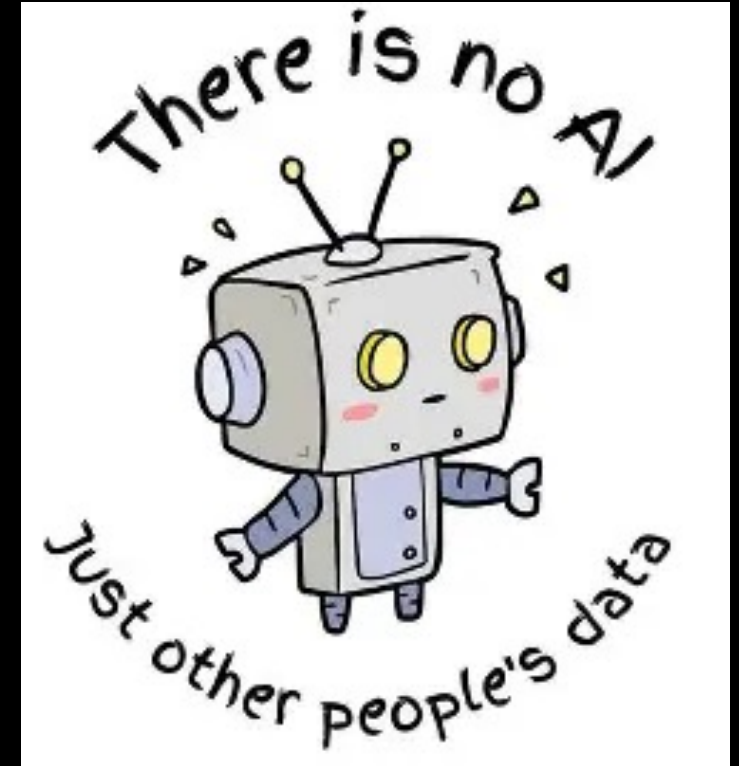
Deliverable 1: Name of the Scientific Figure – due to Jan 28th, 11:59 pm

Deliverable 2: Script – due to Feb 6th, 11:59 pm

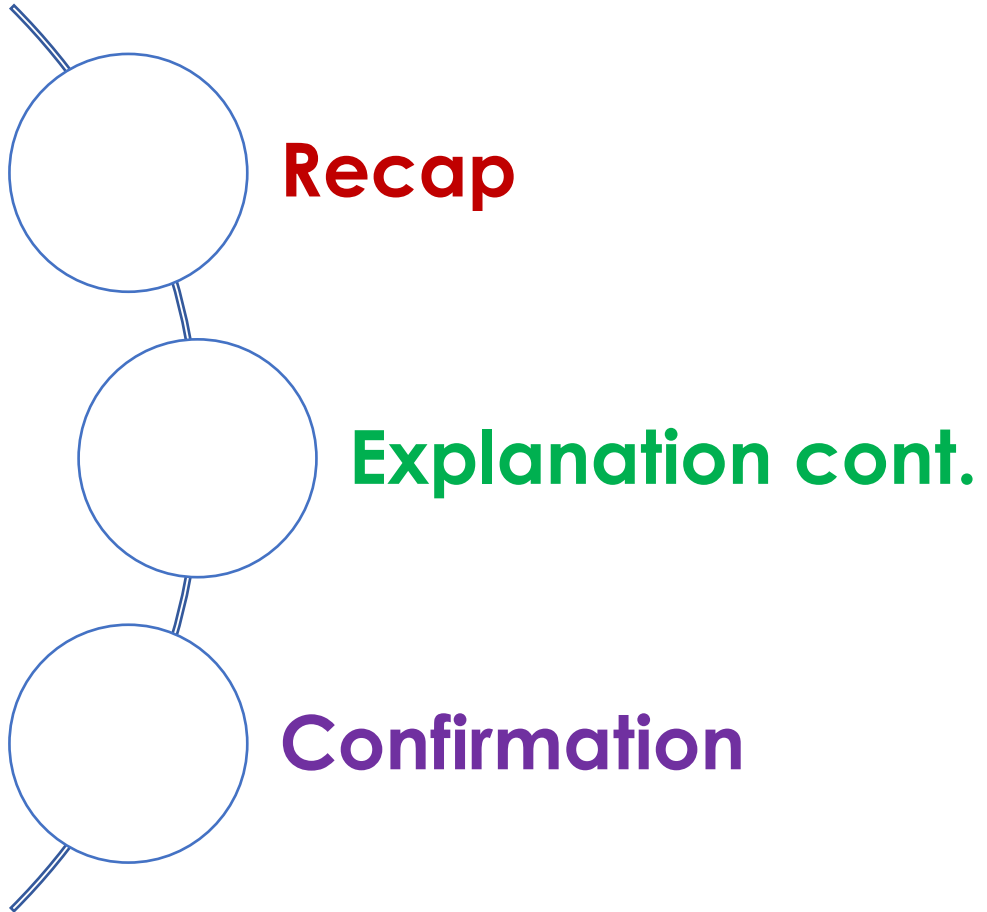
Deliverable 3: Full Submission – due to Feb 15th, 11:59 pm

ANNOUNCEMENT

- **Misuse of AI in Class** - We are Going Back to Pen and Paper!
- I will print you the texts in advance!
- Each group needs to have and to show me a set of notes (every time)!



The Plan!



RECAP



EXPLANATION

- What does it mean to *explain* something to someone (scientifically)?
 - *Why* do objects fall to the earth?
 - *Why* are the leaves of trees green?
 - *Why* does coffee help us wake up?

Relevant Questions:
What are we asking after? What would satisfy us? What wouldn't satisfy us?

EXPLANATION



- What does it mean to *explain* something to someone (scientifically)?
 - Scientific explanations enhance our understanding of the world and are useful as indicators of truth
 - Explanation is a desirable **end**, but it is also a **means** to something else.
 - Similarly, it is a good thing for a theory be explanatory, not just for the explanation's sake but for the truth's sake as well.
 - A theory that explains phenomena is *more likely to be true* than one that doesn't.

Last time!

- What does it explain? However, consider the following examples: the (scientifically)?
 - Scientific explanations of the natural world
 - and are used to explain the world
 - Explanations of the world

The Miasma theory explains the cholera and the plague.

The theory of oceanic sirens explains the tides.

What explains the phenomena is *more likely to be true*



Last time!

Are these theories true?

Nope!

- What does it mean for a theory to be true (scientifically)?
 - Scientific explanations are true if they accurately describe the world
 - and are useful
 - Explanations are useful if they explain something else.
- Similarly, it is useful to have a true theory, not just for the explanation's sake but for the truth's sake as well.
 - A theory that explains phenomena is *more likely to be true* than one that doesn't.

Thus, we might ask:

- What does it mean to explain something (scientifically)?
 - Scientific explanations are the external feature of explaining the world
 - and are used to explain observable phenomena a truth-conducive virtue?
 - Explanations are not something else.
- Similarly, it is not just for the explanation's sake but for the truth's sake as well.
 - A theory that explains phenomena is *more likely to be true* than one that doesn't.

EXPLANATION: THE STANDARD MODEL



Last time!

The Standard Model:

“Any claim to be describing the form of scientific explanation is best understood not as presenting the single form into which all good explanations actually fit but as outlining a paradigm, an exemplar to which most good explanations are similar. It is a core concept with variations and extensions as particular circumstances warrant. It is the central idea from which actual explanations are more or less divergent and about which actual cases are centrally located.

*The standard model of this central idea is called by philosophers of science the **covering-law** model of explanation. Its initial formulation is the work of **Carl Hempel**.”*

EXPLANATION: THE STANDARD MODEL

Last time

- The Standard Model:
 - A phenomenon is explained by demonstration that it is an instance of a law of nature (scientific laws); an event is explained when we articulate the law that covers it.

Example from daily life:

- Someone unfamiliar with traffic asks why the cars are stopping before crossing the intersection. ← event
- *Explanation:* Because there is a stop sign on ther corner, ← the relevant condition of the particular case
and everyone must stop at a stop sign. ← law



EXPLANATION: THE STANDARD MODEL

Model:

is explained
by the law of

Once the law is understood and the conditions are recognized, the event is to be expected .

Had the law and conditions been known, the event could have been *predicted*!

There is a **stricter** connection between the **law** and **event** in question: The statement of the law and the relevant conditions must entail the statement of the event.

"Everyone must stop at the stop sign."

"There is a stop sign."

"We must stop"

This is a **logical requirement!**

- Someone unfamiliar with traffic asks why **the cars are stopping before crossing the intersection.** ← event
- *Explanation:* Because there is **a stop sign on ther corner,** ← the relevant **condition** of the particular case
and ***everyone must stop at a stop sign.*** ← law

EXPLANATION: THE STANDARD MODEL

Last time!

- With this strong connection between law and event, the ***covering-law model*** of explanation is also referred to as the

Deductive-nomological model of explanation (short, the D-N model)

It's a *deductive* argument: if the premises are true, then the conclusion must be true as well.

Nomological: law-governed, a word for “involving laws of nature”.

Side note on terminology



Deductive argument

An argument with the following property: if the premises are true, then the conclusion must be true as well.

Inductive argument:

Non-deductive argument.

Lacks the above property.

Good inductive argument: if the premises are true, then the conclusion is likely true as well.

Side note on terminology



Deductive argument

Dorothy is an engineer. All engineers are good at math. Therefore, Dorothy is good at math.

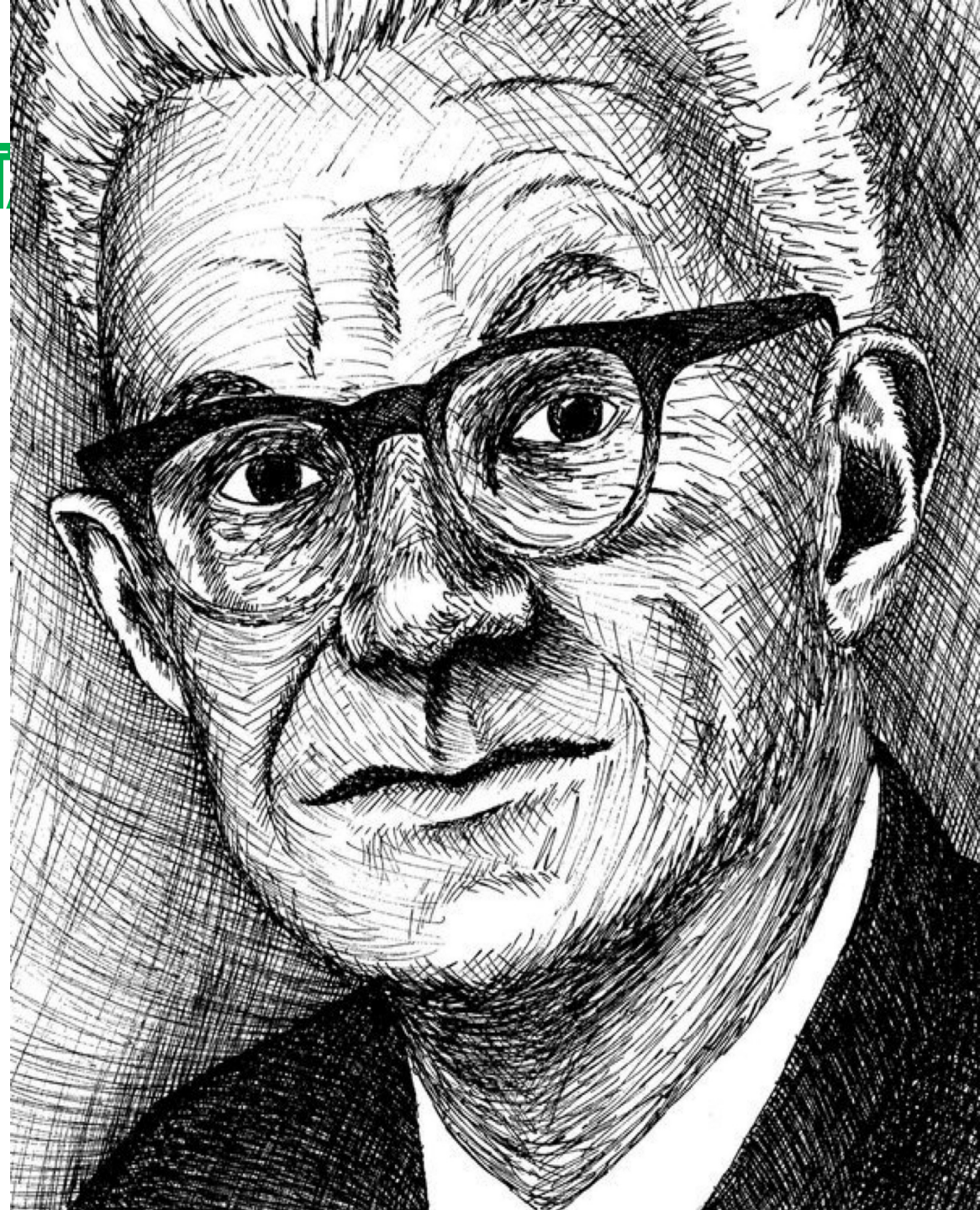
Inductive argument:

Most engineers are good at math. Larry Page (co-founder of Google) is an engineer. Therefore, Larry is probably good at math.



EXPLANATION: THE ST

If law L (or laws $L_1, L_2, \dots$) is true and conditions $C_1, C_2, C_3, \dots$ hold, then event E must be true.





EXPLANATION: THE ST

If law L (or laws $L_1, L_2, \dots$) is true and conditions $C_1, C_2, C_3, \dots$ hold, then event E must be true.

To explain E , we show how to derive it in a logical argument using C s and L s. If the C s and L s are true, then the E must also be true (it is implied, since this is *deduction*).

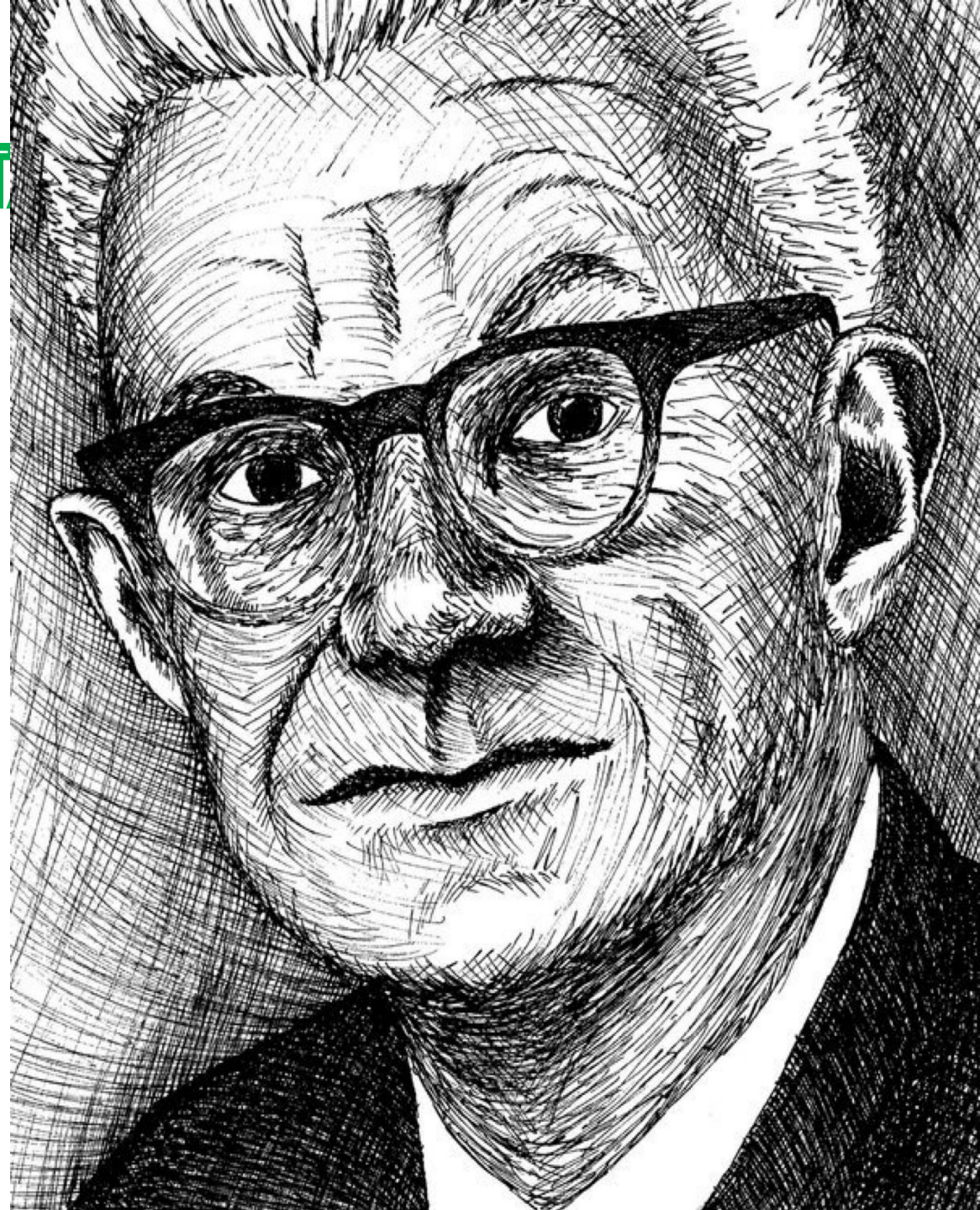




EXPLANATION: THE ST

- **Explanation**

- Ok Hempel,
- *Why* does the spinning speed of the figure skater increase when she pulls her arms in?





EXPLANATION: THE STANDARD MODEL

- If L_1 : The law of conservation of angular momentum
- and*
- C_1 : The skater is rotating
 - C_2 : The skater reduces her moment of inertia by bringing her arms close to her body
 - C_3 : The skater is not interacting with any external body that might alter her angular velocity

then

- E: The skater's rate of rotation increases



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Statements of
initial
conditions

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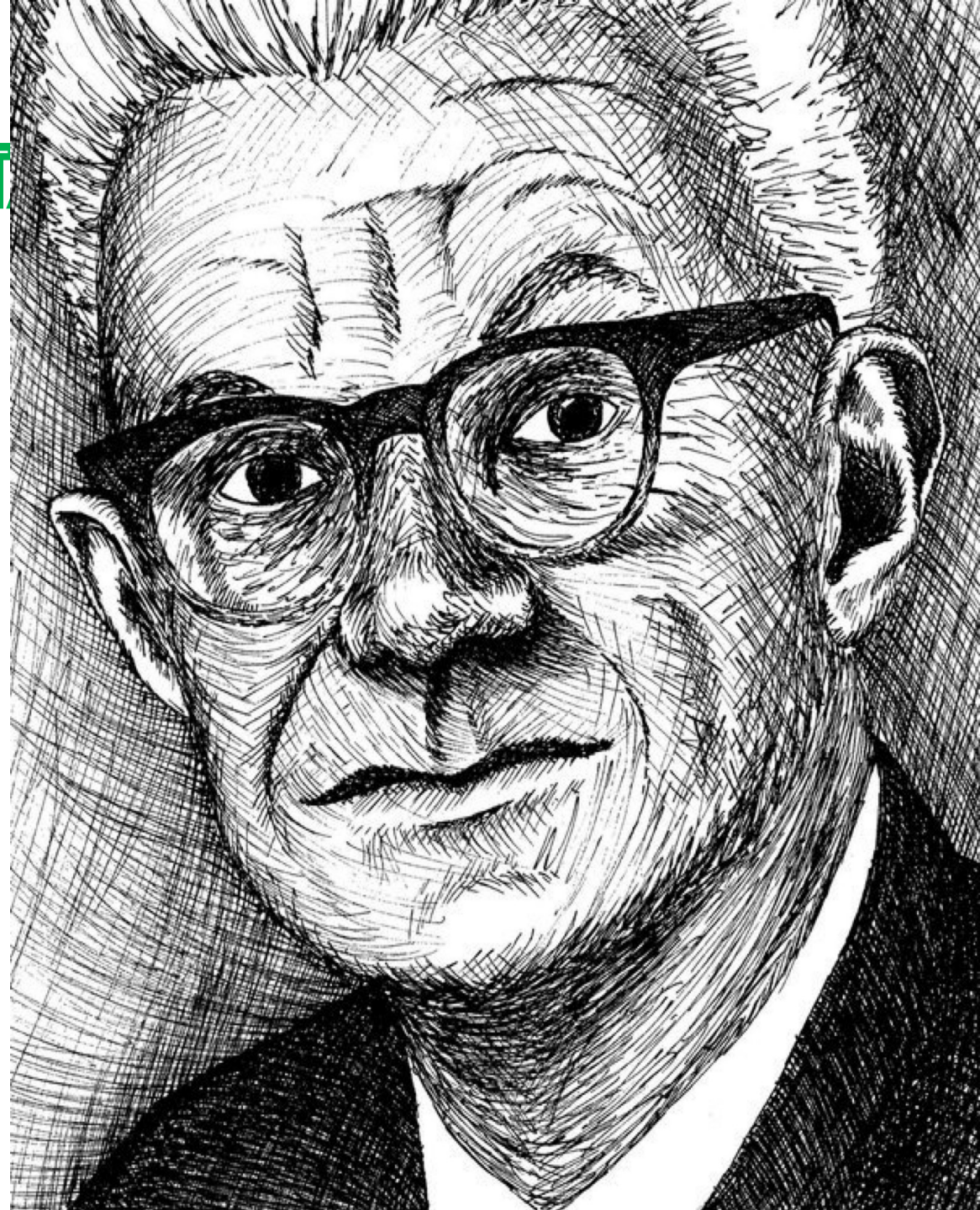


The conclusion of the argument is the event to be explained (Why x?)



EXPLANATION: THE ST

- **Explanation**
 - That's what it means to explain something





EXPLANATION: THE STANDARD MODEL



Reflect & Discuss:

3. Sometimes, the model of explanation that we have just considered **does not work satisfactorily due to a problem of symmetry**. In your reading, this is explained (!) with an example involving the explanation of the mass of the earth. ***“The problem is that the formal analysis is symmetric but explanations are not.”***

Come up with another event whose explanation has the same problem.



EXPLANATION: THE STANDARD MODEL

- Why is there a stop sign at the corner?
- If L_1 : The law that the only place one must stop is at a stop sign
and
- $C1$: We must stop

then

- E: Because we stopped.



EXPLANATION: THE STANDARD MODEL

- Why is there a stop sign at the corner?
- If L: The law that the only place one must stop is at a stop sign

a

- This does not explain why the stop sign is there

then

- E: Because we stopped.

Last time!

EXPLANATION: THE STANDARD MODEL

- Why is there a stop sign at the corner?
- If I... The only place one must stop is at a stop sign

A proper explanation of why the sign is there must cite the city's efforts at traffic safety.

then

- E: Because we stopped.



EXPLANATION: THE STANDARD MODEL

- The *covering-law model* of explanation includes a **problem of distinguishing laws from accidental generalizations**. (**Problem of Relevance**)

HYPOTHESES & LAWS



- Law (Def.):

- A claim that associates a general kind of thing with a particular behavior or property is a law.
- Laws identify natural kinds in the world and imply a **causal connection** between being that kind of thing and having the associated property or behavior.
- Claims about accidental associations and the circumstantial properties of such groups are not laws, even though they are generalizations.
- ***Laws support prediction; accidental generalization do not.***

EXPLANATION: THE STANDARD MODEL



Last time!

- Determination of what counts as law will depend on our *theoretical background* to guide the *decision* as to *what is relevant and what is not*.
- The theories we hold, then, influence what counts as an explanation.
- An evaluation of this external virtue, that is, will be influenced by an internal reference to other theories.

EXPLANATION CONT.

EXPLANATION: THE STANDARD MODEL

- The **covering-law model** of explanation has ignored several important issues regarding its **adequacy**:
 - For example, cases of *statistical laws* in which the relevant conditions don't always bring about (cause) the phenomena.
 - Example: Sarah's death is explained by her having leukemia. But not every case of leukemia is fatal, so why did it happen **this** time?

EXPLANATION: THE STANDARD MODEL

- This brief account is sufficient to raise the main issue of explanation as a truth-conducive virtue of theories.
- If being a good explanation of phenomena is to function as a truth-conducive virtue, it must be a feature we can evaluate.
- So, what *is* a good explanation?

EXPLANATION: THE STANDARD MODEL

- I. So far, we've seen that this feature (i.e., the truth-conducive virtue of explanation) cannot be evaluated without ***reference to some theories***.
- II. It must be clear that the law and the conditions are *describing relevant factors* and not just circumstantial properties.

AMENDMENTS TO THE COVERING-LAW MODEL



Reflect & Discuss:

“The more fruitful the explanation, the better the explanation.”

- **Explain**

AMENDMENTS TO THE COVERING-LAW MODEL

- An explanation must reveal a ***unity in nature***, and it must be ***fruitful*** in the sense of covering a ***maximum number of phenomena*** with a ***minimum number of laws***.
 - When there are competing explanations, one should always opt for the more fruitful one.
 - This will help with distinguishing true generalizations which, with true conditions, entail the event in question, from accidental (and irrelevant) associations.

AMENDMENTS TO THE COVERING-LAW MODEL



Reflect & Discuss:

“Successful explanation certainly doesn’t guarantee that the explaining theory is true. The fact that a theory can explain phenomena does not deductively prove that a theory is true. What then is deductive about the deductive-nomological (D-N) form of explanation?”

- Answer the question posed in the above quote.
- Read excerpt 1, make notes of any important points and definitions.

AMENDMENTS TO THE COVERING-LAW MODEL

- It is the entailment from law (and conditions) to the event:
 - *If* the explaining theory is true (and the conditions hold), then the event in question must happen.
 - *Assuming* the theory is true, the phenomenon is to be *expected*.

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This is **not a deductive proof** that the theory is true, and to argue as if it is is to commit a logical fallacy (error in reasoning)

AMENDMENTS TO THE COVERING

- It is the entailment from law (a) to event (b) that is the covering law.
 - *If* the explaining theory is true, the question must happen.
 - *Assuming* the theory is true, the

An argument of the form “the law (and conditions) entails the event” and “the event is real” so “the law must be true” is a fallacious argument.

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Note:

The law being true is a **sufficient** explanation of the observed event, but *it is not necessary*.

There may be other ways to explain it, and these have not been ruled out.

AMENDMENTS TO THE

The alleged deductive proof of the theory is just as flawed as arguing that whenever the day is the fourth of July it must be summertime, and it is summertime so it must be the fourth of July.

However, it also could be August.

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- Explanation is not a deductive proof that the explanatory theory is true, but it might contribute to an ***inductive*** proof.
 - The *difference* between ***deductive*** and ***inductive*** proofs is exactly the *difference* between ***certainty*** and ***likelihood***.

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 - The *difference* between ***deductive*** and ***inductive*** proofs is exactly the *difference* between ***certainty*** and ***likelihood***.
- A *successful* explanation *increases* the ***likelihood*** that the explanatory theory is true.
- This would make explanation a truth-conducive virtue, if it can be demonstrated.

AMENDMENTS TO THE COVERING-LAW MODEL



Reflect & Discuss

“A good explanation makes good sense out of what had been puzzling. It is, as C. S. Peirce put it, “feeling the key turn in the lock.” All of these criteria of good explanation, this feeling, things making sense, expectation, are psychological factors that depend on a person’s interests, antecedent beliefs, and psychological preparation. None of them really depends on the truth. That satisfying feeling of understanding can come from a false belief as easily as from true belief.”

- **What can make explanations truth-conducive and what is the challenge in doing so.**

AMENDMENTS TO THE COVERING-LAW MODEL

- A theory can be explanatory without being true.
 - Example:
 - The explanation of my neighbor stealing the newspaper might make sense, but it does not mean that it is true.
 - Nonetheless, if I continue to think as if the theory is true, I continue to make sense of the phenomena in my world. Now that I understand the phenomenon, I can deal with it (e.g., I know the newspaper will not be there. Thus, I need to go and buy a new one).
 - **The explanation fully serves its purpose even though its theory is false.**

AMENDMENTS TO THE COVERING-LAW MODEL

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 - On the other hand, *nothing can be the cause of an event unless that thing really exists*, that is, *unless the theory about it as a cause is true*.

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 - On the other hand, *nothing can be the cause of an event unless that thing really exists*, that is, *unless the theory about it as a cause is true*.
 - ***Acceptance of a causal account as explanation of a phenomenon carries with it a commitment to the truth of the causal theory.***

AMENDMENTS TO THE COVERING-LAW MODEL

- Things are different if the **causal factor** is added to the model of explanation.
 - If you know that the theory explains the phenomenon in the sense of showing that it is to be *expected*, *you don't know that the theory is true*.
 - But if you know that some particular thing *caused* the phenomenon, then you also know that this thing really exists.

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- ***How to evaluate this?***

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 - But if you know that some particular thing *caused* the phenomenon, then you also know that this thing really exists.
- ***How to evaluate this?***
 - *The causality and success of an explanation are evaluated by **comparison to theories**, and **explanation as an external virtue**, as a form of comparison between theory and data.*